BİL4010

Paralel programlama

13.Hafta

MPI ile Matris-Vektör Çarpımı

İçerik

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Matris Tanımlanması

Definition A matrix is a rectangular array of numbers.

$$A = \underbrace{\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & a_{ij} & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}}_{n \text{ columns}} \left.\vphantom{\begin{bmatrix} a \\ a \\ a \\ a \end{bmatrix}}\right\} m \text{ rows}$$

element in ith row, jth column

Also written as $A=[a_{ij}]$

$m \times n$ matrix

When $m=n$, A is called a square matrix.

Matris Toplama/Çıkarma

Let $A = [a_{ij}]$, $B = [b_{ij}]$ be $m \times n$ matrices. Then:
$A + B = [a_{ij} + b_{ij}]$, and $A - B = [a_{ij} - b_{ij}]$

$$\begin{bmatrix} 1 & -1 \\ 3 & 4 \\ 2 & 0 \end{bmatrix} + \begin{bmatrix} 3 & 4 \\ 1 & -4 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 3 \\ 4 & 0 \\ 4 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 \\ 3 & 4 \\ 2 & 0 \end{bmatrix} - \begin{bmatrix} 3 & 4 \\ 1 & -4 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} -2 & -5 \\ 2 & 8 \\ 0 & -3 \end{bmatrix}$$

Matris Tersi

- A, nxn boyutunda kare bir matris olsun,
- A matrisinin tersi A^{-1} olarak gösterilir.
- A matrisi yalnızca $A \times A^{-1} = I$ olduğunda tersine çevrilebilir.
  - $I_n = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
- Bir sayının sayısal tersi ile benzer ancak aynı değil: 1/8 = 0,125
- 6 * (1/6) ile benzer
- Kare matrislerin tamamı tersine çevrilemezler!

Ters: AX = K şeklindeki Denklemleri Çözmek için Kullanılır

$$\begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

$$\begin{bmatrix} a_1x + a_2y + a_3z \\ b_1x + b_2y + b_3z \\ c_1x + c_2y + c_3z \end{bmatrix} = \begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

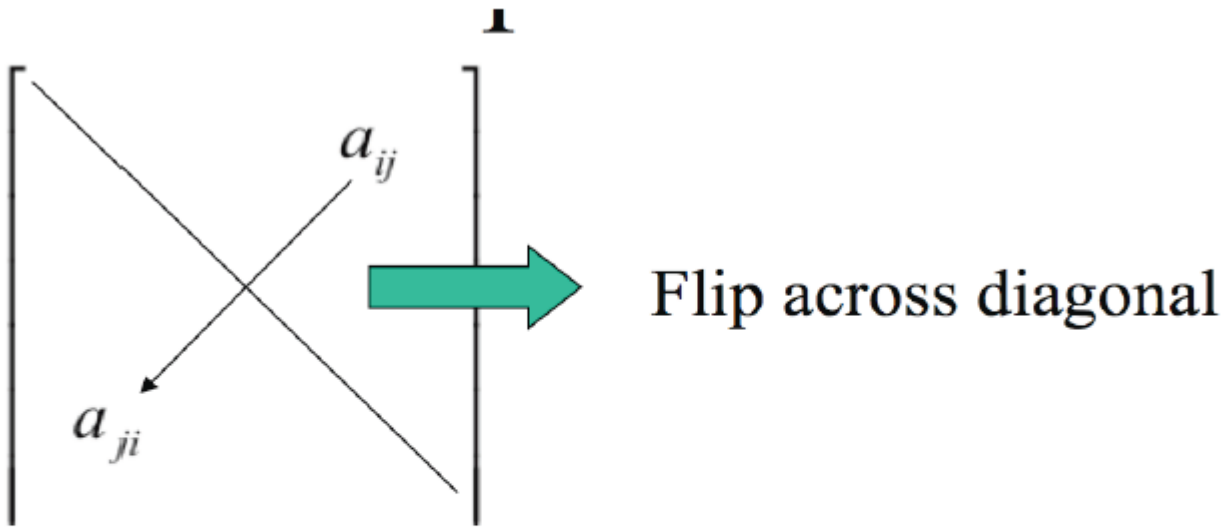
$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} a^{-1}{}_1 & a^{-1}{}_2 & a^{-1}{}_3 \\ b^{-1}{}_1 & b^{-1}{}_2 & b^{-1}{}_3 \\ c^{-1}{}_1 & c^{-1}{}_2 & c^{-1}{}_3 \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

$$AX = K$$

$$A^{-1}AX = A^{-1}K$$

$$IX = A^{-1}K$$

Seri Matris Transpozu



$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

written A^t

Anapara ve Faizin Hesaplanması

- Üyeler: Banka müşterileri
  - Her üyenin ID'sine göre bir kullanıcı adı vardır
  - Üye ID, üye hesapları tablosundaki üyeler satırına eşittir
- Hesaplar: N hesap türü vardır:
  - kontrol etme; kolej (tasarruf); tatil (tasarruf); CD; IRA (yatırım).
  - Tüm üyelerin tüm hesaplarında parası olmayabilir (seyrek veri).
  - Hesaplar faizlidir:
    - Her hesabın kendi faiz oranı vardır: r_n
    - Tüm oranların pozitif olduğunu varsayalım
    - Tahakkuk eden Aylık Faiz Oranı: $R[r_1; r_2; \ldots; r_n]$
- Hesap Matrisi (A), tüm üyelerin (satır) hesap bakiyelerini saklar (sütunlar)
- *Her Üyenin (i)* faiz getiren hesap bakiyelerinin toplamına eşit bir toplam hesap bakiyesi vardır:

$$Acct_{total}(i) = \sum_{j=1}^{n} a_{ij} = a_{i1} + \cdots + a_{ij} + \cdots + a_{in}$$

Matris-Vektör Çarpımı: Anapara ve Faiz hesabı

- Ay sonunda her üye, her hesaptaki bakiyeye göre faiz kazanır.
- Her üye hesabında kazanılan toplam faiz şu şekilde hesaplanır:

$$B_{int} = A * R, \text{ where}$$

$$B_{int} = \begin{bmatrix} b_1 \\ b_2 \\ \dots \\ b_i \\ \dots \\ b_m \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1j} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2j} & \dots & a_{2n} \\ & & \dots & & & \\ a_{i1} & a_{i2} & \dots & a_{ij} & \dots & a_{in} \\ & & \dots & & & \\ a_{m1} & a_{m2} & \dots & a_{mj} & \dots & a_{mn} \end{bmatrix} * \begin{bmatrix} r_1 \\ r_2 \\ \dots \\ r_i \\ \dots \\ r_n \end{bmatrix}$$

$$b_i = \sum_{j=1}^{n} a_{ji} = a_{i1} * r_1 + \cdots + a_{ij} * r_j + \cdots + a_{in} * r_n$$

- Üyelerin yeni toplam hesap bakiyeleri;

$$Acct_{new} = Acct_{total} + B_{int}$$

Matris-Vektör Çarpımı

$A = (a_{ij})$ is an $m \times n$ matrix

$\mathbf{x}$ is a vector with n components

$\mathbf{y} = A\mathbf{x}$ is a vector with m components

$$y_i = a_{i0}x_0 + a_{i1}x_1 + a_{i2}x_2 + \cdots a_{i,n-1}x_{n-1}$$

i-th component of y

Dot product of the ith row of A with x.

Matris-Vektör Çarpımı

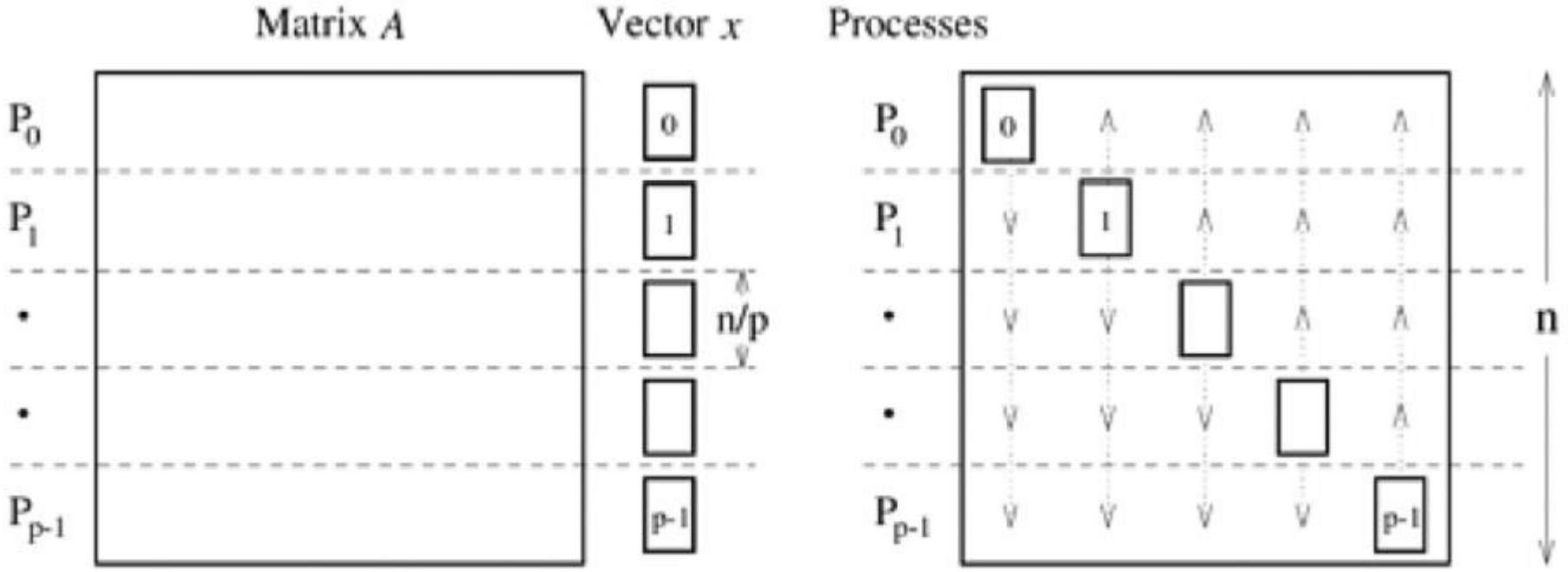
$$
\begin{bmatrix}
a_{00} & a_{01} & \cdots & a_{0,n-1} \\
a_{10} & a_{11} & \cdots & a_{1,n-1} \\
\vdots & \vdots & & \vdots \\
a_{i0} & a_{i1} & \cdots & a_{i,n-1} \\
\vdots & \vdots & & \vdots \\
a_{m-1,0} & a_{m-1,1} & \cdots & a_{m-1,n-1}
\end{bmatrix}
\begin{bmatrix}
x_0 \\ x_1 \\ \vdots \\ x_{n-1}
\end{bmatrix}
=
\begin{bmatrix}
y_0 \\ y_1 \\ \vdots \\ y_i = a_{i0}x_0 + a_{i1}x_1 + \cdots a_{i,n-1}x_{n-1} \\ \vdots \\ y_{m-1}
\end{bmatrix}
$$

Matris-Vektör Çarpımı – Seri Pseudo Code

```
/* For each row of A */
for (i = 0; i < m; i++) {
    /* Form dot product of ith row with x */
    y[i] = 0.0;
    for (j = 0; j < n; j++)
        y[i] += A[i][j]*x[j];
}
```

Matris-Vektör Çarpma Bölümleme: 1-D Rowwise

- *nxn* matrisi, *p* işlemci arasında bölünmüştür; her işlemci, matrisin n/p satırını depolar.
- *nx1* vektörü *x*, her işlem kendi elemanlarının n/p'sine sahip olacak şekilde dağıtılır.

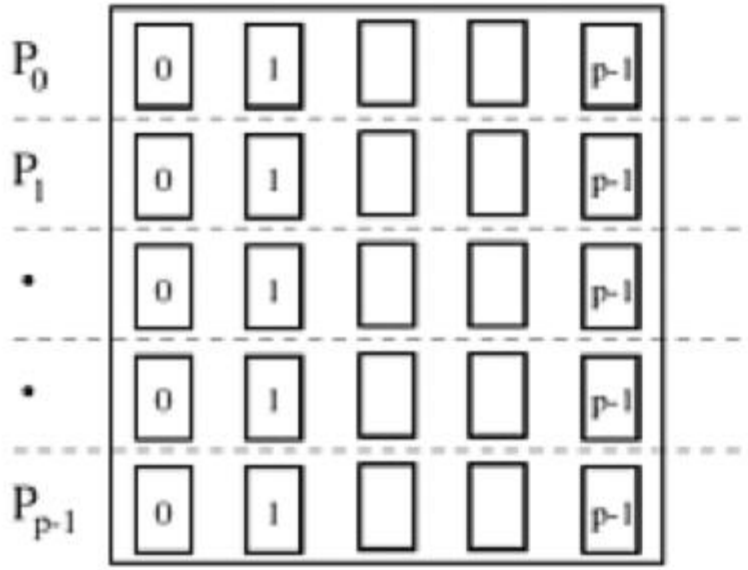


(a) Initial partitioning of the matrix and the starting vector *x*

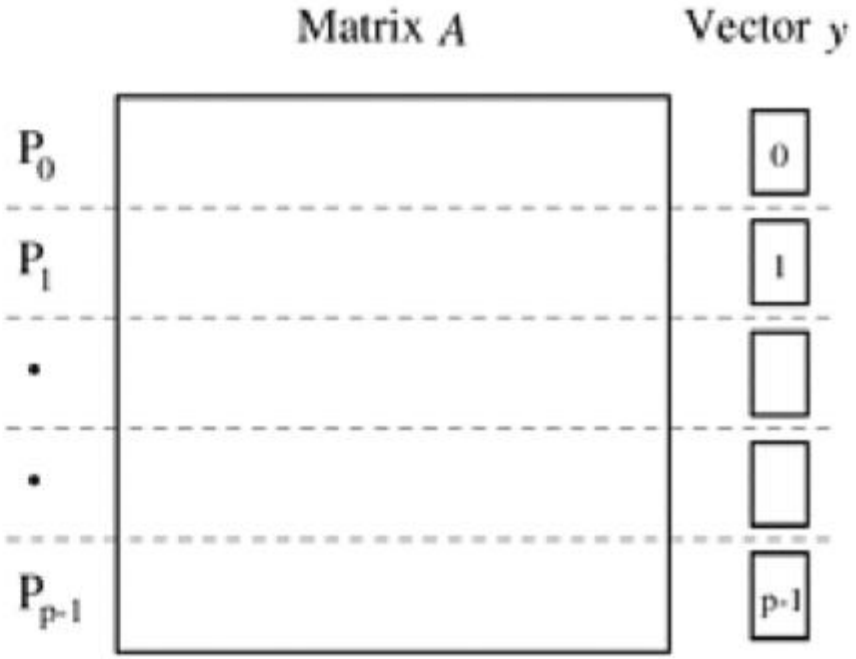
(b) Distribution of the full vector among all the processes by all-to-all broadcast

Matris-Vektör Çarpma Bölümleme: 1-D Rowwise

- *x* vektörünün all-to-all biçimde paylaşılması gereklidir.
- *n x 1* vektörü *x*, her işlem kendi elemanlarının n/p'sine sahip olacak şekilde dağıtılır.



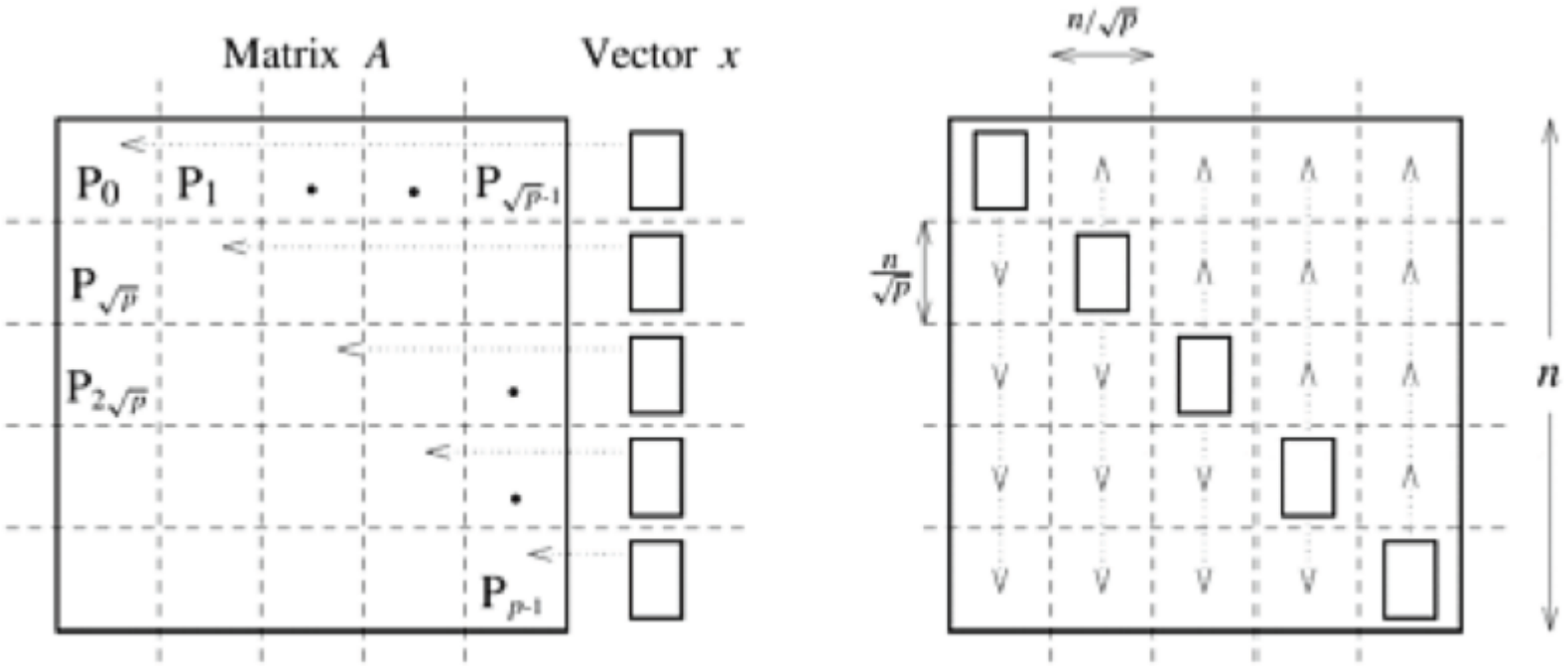
(c) Entire vector distributed to each process after the broadcast



(d) Final distribution of the matrix and the result vector *y*

Matris-Vektör Çarpma Bölümleme: 2-B Blok-Blok

- $n \times n$ matrisi, p işlemci arasında bölünür; her işlemci, matrisin $n/\sqrt{p}$ satır x sütununu depolar.
- $n \times 1$ vektörü x öyle dağıtılır ki, her satırdaki işlemler $n/\sqrt{p}$ elemanına sahip olur.

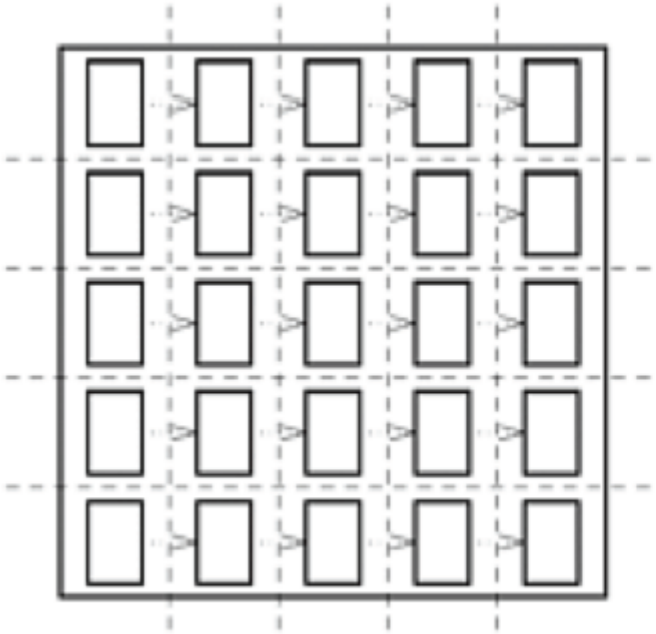


(a) Initial data distribution and communication steps to align the vector along the diagonal

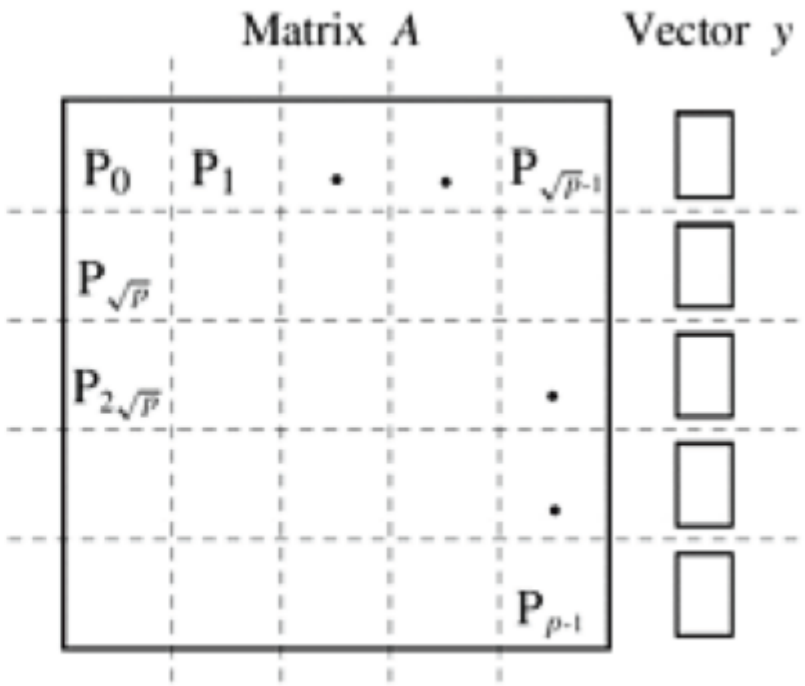
(b) One-to-all broadcast of portions of the vector along process columns

Matris-Vektör Çarpma Bölümleme: 2-B Blok-Blok

- *x* vektörünün all-to-all biçimde paylaşılması gereklidir.
- *n x 1* vektörü *x*, her işlem kendi elemanlarının n/p'sine sahip olacak şekilde dağıtılır.



(c) All-to-one reduction of partial results



(d) Final distribution of the result vector

Bir matrisinin saklanması

$$\begin{pmatrix} 0 & 1 & 2 & 3 \\ 4 & 5 & 6 & 7 \\ 8 & 9 & 10 & 11 \end{pmatrix}$$

stored as

0 1 2 3 4 5 6 7 8 9 10 11

Seri Matris-Vektör Çarpımı

```
void Mat_vect_mult(
        double  A[]  /* in  */,
        double  x[]  /* in  */,
        double  y[]  /* out */,
        int     m    /* in  */,
        int     n    /* in  */) {
    int i, j;

    for (i = 0; i < m; i++) {
        y[i] = 0.0;
        for (j = 0; j < n; j++)
            y[i] += A[i*n+j]*x[j];
    }
}  /* Mat_vect_mult */
```

MPI ile Matris-Vektör Çarpımı

```
void Mat_vect_mult(
      double    local_A[]  /* in  */,
      double    local_x[]  /* in  */,
      double    local_y[]  /* out */,
      int       local_m    /* in  */,
      int       n          /* in  */,
      int       local_n    /* in  */,
      MPI_Comm  comm       /* in  */) {
   double* x;
   int local_i, j;
   int local_ok = 1;


   x = malloc(n*sizeof(double));
   MPI_Allgather(local_x, local_n, MPI_DOUBLE,
         x, local_n, MPI_DOUBLE, comm);

   for (local_i = 0; local_i < local_m; local_i++) {
      local_y[local_i] = 0.0;
      for (j = 0; j < n; j++)
         local_y[local_i] += local_A[local_i*n+j]*x[j];
   }
   free(x);
}  /* Mat_vect_mult */
```

Örn. MPI ile Matris-Vektör Çarpımı

```c
/* File:     mpi_mat_vect_mult.c
 * Purpose:  Implement parallel matrix-vector
 * one-dimensional arrays to store the vectors and the
 * matrix.  Vectors use block distributions and the
 * matrix is distributed by block rows.
  * IPP:      Section 3.4.9 (pp. 113 and ff.)
 */
#include <stdio.h>
#include <stdlib.h>
#include <mpi.h>

void Check_for_error(int local_ok,char fname[],char message[],
      MPI_Comm comm);
void Get_dims(int* m_p,int* local_m_p,int* n_p,int* local_n_p,
      int my_rank,int comm_sz,MPI_Comm comm);
void Allocate_arrays(double** local_A_pp,double** local_x_pp,
      double** local_y_pp,int local_m,int n,int local_n,
      MPI_Comm comm);
void Read_matrix(char prompt[],double local_A[],int m,
     int local_m, int n,int my_rank,MPI_Comm comm);
void Read_vector(char prompt[],double local_vec[],int n,
      int local_n. int my_rank,MPI_Comm comm);
void Print_matrix(char title[],double local_A[],int m,
      int local_m, int n,int my_rank,MPI_Comm comm);
void Print_vector(char title[],double local_vec[],int n,
      int local_n,int my_rank,MPI_Comm comm);
void Mat_vect_mult(double local_A[],double local_x[],
      double local_y[],int local_m,int n,int local_n,
      MPI_Comm comm);
```

```c
/*-------------------------------------------------*/
int main(void) {
   double* local_A, local_x,  local_y;
   int m,local_m,n,local_n, my_rank,comm_sz;
   MPI_Comm comm;

   MPI_Init(NULL,NULL);
   comm = MPI_COMM_WORLD;
   MPI_Comm_size(comm,&comm_sz);
   MPI_Comm_rank(comm,&my_rank);

   Get_dims(&m,&local_m,&n,&local_n,my_rank,
         comm_sz,comm);
   Allocate_arrays(&local_A,&local_x,&local_y,local_m,
             n,local_n,comm);
   Read_matrix("A",local_A,m,local_m,n,
              my_rank,comm);
#  ifdef DEBUG
   Print_matrix("A",local_A,m,local_m,n,my_rank,comm);
#  endif
   Read_vector("x",local_x,n,local_n,my_rank,comm);
#  ifdef DEBUG
   Print_vector("x",local_x,n,local_n,my_rank,comm);
#  endif

   Mat_vect_mult(local_A,local_x,local_y,local_m,n,local_n,comm);

   Print_vector("y",local_y,m,local_m,my_rank,comm);

   free(local_A); free(local_x); free(local_y);
   MPI_Finalize();
   return 0;
}  /* main */
```

Örn. MPI ile Matris-Vektör Çarpımı

```
/* PACHECO MAT-VEC-MULT (cont.) */
* Function:  Get_dims
 * Purpose:   Get the dimensions of the matrix and the vectors from stdin.
 * In args:   my_rank:   calling processes rank in comm
 *            comm_sz:   number of processes in comm
 *            comm:      communicator containing all processes calling Get_dims
 * Out args:  m_p:       global number of rows of A and components in y
 *            local_m_p: local number of rows of A and components of y
 *            n_p:       global number of cols of A and components of x
 *            local_n_p: local number of components of x
 *
 * Errors:    if either m or n isn't positive or if m or n isn't evenly  divisible by comm_sz,the program
 *            prints an error message and quits.
 * Note:
 *    All processes in comm should call Get_dims
 */
void Get_dims(
      int*      m_p        /* out */,
      int*      local_m_p  /* out */,
      int*      n_p        /* out */,
      int*      local_n_p  /* out */,
      int       my_rank    /* in  */,
      int       comm_sz    /* in  */,
      MPI_Comm  comm       /* in  */) {
      int local_ok = 1;

   if (my_rank == 0) {
      printf("Enter the number of rows\n");        scanf("%d",m_p);
      printf("Enter the number of columns\n");      scanf("%d",n_p);          }

   MPI_Bcast(m_p,1,MPI_INT,0,comm);
   MPI_Bcast(n_p,1,MPI_INT,0,comm);
   if (*m_p <= 0 || *n_p <= 0 || *m_p % comm_sz != 0
         || *n_p % comm_sz != 0) local_ok = 0;
   Check_for_error(local_ok,"Get_dims", "m and n must be positive and evenly divisible by comm_sz", comm);

   *local_m_p = *m_p/comm_sz;
   *local_n_p = *n_p/comm_sz;
}  /* Get_dims */
```

Örn. MPI ile Matris-Vektör Çarpımı

```
/* PACHECO MAT-VEC-MULT (cont.) */
/*-------------------------------------------------------------------
 * Function:  Check_for_error
 * Purpose:   Check whether any process has found an error.  If so, print message and terminate all processes.
 *                  Otherwise, continue execution.
 * In args:   local_ok:  1 if calling process has found an error, otherwise
 *            fname:     name of function calling Check_for_error
 *            message:   message to print if there's an error
 *            comm:      communicator containing processes calling
 *                       Check_for_error:  should be MPI_COMM_WORLD.
 *
 * Note:
 *    The communicator containing the processes calling Check_for_error
 *    should be MPI_COMM_WORLD.
 */
void Check_for_error(
      int       local_ok   /* in */,
      char      fname[]    /* in */,
      char      message[]  /* in */,
      MPI_Comm  comm       /* in */) {
   int ok;

   MPI_Allreduce(&local_ok,&ok,1,MPI_INT,MPI_MIN,comm);
   if (ok == 0) {
      int my_rank;
      MPI_Comm_rank(comm,&my_rank);
      if (my_rank == 0) {
         fprintf(stderr,"Proc %d > In %s,%s\n",my_rank,fname,
               message);
         fflush(stderr);
      }
      MPI_Finalize();
      exit(-1);
   }
}  /* Check_for_error */
```

Örn. MPI ile Matris-Vektör Çarpımı

```
/* PACHECO MAT-VEC-MULT (cont.) */
/*-------------------------------------------------------------------
 * Function:   Allocate_arrays
 * Purpose:    Allocate storage for local parts of A,x,and y
 * In args:    local_m:    local number of rows of A and components of y
 *             n:          global and local number of cols of A and global
 *                         number of components of x
 *             local_n:    local number of components of x
 *             comm:       communicator containing all calling processes
 * Out args:   local_A_pp: local storage for matrix (m/comm_sz rows,n cols)
 *             local_x_pp: local storage for x (n/comm_sz components)
 *             local_y_pp: local_storage for y (m/comm_sz components)
 *
 * Errors:     if a malloc fails,the program prints a message and all
 *             processes quit
 * Note:
 *    Communicator should be MPI_COMM_WORLD because of call to
 * Check_for_errors
 */
```

```
void Allocate_arrays(
      double**  local_A_pp  /* out */,
      double**  local_x_pp  /* out */,
      double**  local_y_pp  /* out */,
      int       local_m     /* in  */,
      int       n           /* in  */,
      int       local_n     /* in  */,
      MPI_Comm  comm        /* in  */) {

   int local_ok = 1;

   *local_A_pp = malloc(local_m*n*sizeof(double));
   *local_x_pp = malloc(local_n*sizeof(double));
   *local_y_pp = malloc(local_m*sizeof(double));

   if (*local_A_pp == NULL || local_x_pp == NULL ||
         local_y_pp == NULL) local_ok = 0;
   Check_for_error(local_ok,"Allocate_arrays",
         "Can't allocate local arrays",comm);
}  /* Allocate_arrays */
```

Örn. MPI ile Matris-Vektör Çarpımı

```
/*-----------------------------------------------------
 * Function:  Read_matrix
 * Purpose:   Read in the matrix and distribute among
 *           the processes using a block row distribution
 * In args:   prompt:  description of matrix (e.g.,"A")
 *            m:       global number of rows of A
 *            local_m: local number of rows of A
 *            n:       global and local number of cols of A
 *            my_rank: process rank in communicator comm
 *            comm:    communicator containing processes
 *                     calling Read_matrix
 * Out args:  local_A: the local matrix
 *
 * Errors:    if malloc of temporary storage fails on process
 *           0,the program prints a message and all processes quit
 * Note:
 * 1. Communicator should be MPI_COMM_WORLD
 *      because of call to Check_for_errors
 * 2. local_m and n should be the same on each process
 */
void Read_matrix(
      char      prompt[]   /* in  */,
      double    local_A[]  /* out */,
      int       m          /* in  */,
      int       local_m    /* in  */,
      int       n          /* in  */,
      int       my_rank    /* in  */,
      MPI_Comm  comm       /* in  */) {
   double* A = NULL;
   int local_ok = 1;
   int i,j;
```

```
if (my_rank == 0) {
   A = malloc(m*n*sizeof(double));
   if (A == NULL) local_ok = 0;
   Check_for_error(local_ok,"Read_matrix",
         "Can't allocate temporary matrix",comm);
   printf("Enter the matrix %s\n",prompt);
   for (i = 0; i < m; i++)
      for (j = 0; j < n; j++)
         scanf("%lf",&A[i*n+j]);
   MPI_Scatter(A,local_m*n,MPI_DOUBLE,
         local_A,local_m*n,MPI_DOUBLE,0,comm);
   free(A);
} else {
   Check_for_error(local_ok,"Read_matrix",
         "Can't allocate temporary matrix",comm);
   MPI_Scatter(A,local_m*n,MPI_DOUBLE,
         local_A,local_m*n,MPI_DOUBLE,0,comm);
}
}  /* Read_matrix */
```

Örn. MPI ile Matris-Vektör Çarpımı

```
/*-----------------------------------------------------------
 * Function:  Read_vector
 * Purpose:   Read a vector from stdin and distribute among the
 *            processes using a block distribution
 * In args:   prompt:  description of vector (e.g.,"x")
 *            n:       global order of vector
 *            local_n: local order of vector (n/comm_sz)
 *
 * Errors:    if malloc of temporary storage fails on process
 *          0, program prints a message and all processes quit
 * Notes:
 * 1. Communicator should be MPI_COMM_WORLD
 *        because of call to  Check_for_errors
 * 2. local_n should be the same on all processes
 */
void Read_vector(
      char      prompt[]     /* in  */,
      double    local_vec[]  /* out */,
      int       n            /* in  */,
      int       local_n      /* in  */,
      int       my_rank      /* in  */,
      MPI_Comm  comm         /* in  */) {
   double* vec = NULL;
   int i,local_ok = 1;
   if (my_rank == 0) {
      vec = malloc(n*sizeof(double));
      if (vec == NULL) local_ok = 0;
      Check_for_error(local_ok,"Read_vector",
            "Can't allocate temporary vector",comm);
      printf("Enter the vector %s\n",prompt);
      for (i = 0; i < n; i++)
         scanf("%lf",&vec[i]);
      MPI_Scatter(vec,local_n,MPI_DOUBLE,
            local_vec,local_n,MPI_DOUBLE,0,comm);
      free(vec);
   } else {
      Check_for_error(local_ok,"Read_vector",
            "Can't allocate temporary vector",comm);
      MPI_Scatter(vec,local_n,MPI_DOUBLE,
            local_vec,local_n,MPI_DOUBLE,0,comm);
   }
}  /* Read_vector */
```

Örn. MPI ile Matris-Vektör Çarpımı

```
/*-----------------------------------------------------------
 * Function:  Print_matrix
 * Purpose:   Print a matrix distributed by block rows to stdout
 * In args:   title:    name of matrix
 *            local_A:  calling process' part of matrix
 *            m:        global number of rows
 *            local_m:  local number of rows (m/comm_sz)
 *            n:        global (and local) number of cols
 *            my_rank:  calling process' rank in comm
 *            comm:     communicator containing all processes
 * Errors:    if malloc of local storage on process 0 fails,all
 *            processes quit.
 * Notes:
 * 1.  comm should be MPI_COMM_WORLD because of call
 *      to Check_for_errors
 * 2.  local_m should be the same on all the processes
 */
void Print_matrix(
      char      title[]    /* in */,
      double    local_A[]  /* in */,
      int       m          /* in */,
      int       local_m    /* in */,
      int       n          /* in */,
      int       my_rank    /* in */,
      MPI_Comm  comm       /* in */) {
   double* A = NULL;
   int i,j,local_ok = 1;
```

```
   if (my_rank == 0) {
      A = malloc(m*n*sizeof(double));
      if (A == NULL) local_ok = 0;
      Check_for_error(local_ok,"Print_matrix",
            "Can't allocate temporary matrix",comm);
      MPI_Gather(local_A,local_m*n,MPI_DOUBLE,
            A,local_m*n,MPI_DOUBLE,0,comm);
      printf("\nThe matrix %s\n",title);
      for (i = 0; i < m; i++) {
         for (j = 0; j < n; j++)
            printf("%f ",A[i*n+j]);
         printf("\n");
      }
      printf("\n");
      free(A);
   } else {
      Check_for_error(local_ok,"Print_matrix",
            "Can't allocate temporary matrix",comm);
      MPI_Gather(local_A,local_m*n,MPI_DOUBLE,
            A,local_m*n,MPI_DOUBLE,0,comm);
   }
}  /* Print_matrix */
```

Örn. MPI ile Matris-Vektör Çarpımı

```
/*-----------------------------------------------------------------
 * Function:  Print_vector
 * Purpose:   Print a vector with a block distribution
 * In args:   title:      name of vector
 *            local_vec:  calling process' part of vector
 *            n:          global number of components
 *            local_n:    local number of components (n/comm_sz)
 *            my_rank:    calling process' rank in comm
 *            comm:       communicator containing all processes
 * Errors:    if malloc of local storage on process 0 fails,all
 *            processes quit.
 * Notes:
 * 1.  comm should be MPI_COMM_WORLD because of call
 *             to Check_for_errors
 * 2.  local_n should be the same on all the processes
 */
void Print_vector(
      char      title[]     /* in */,
      double    local_vec[] /* in */,
      int       n           /* in */,
      int       local_n     /* in */,
      int       my_rank     /* in */,
      MPI_Comm  comm        /* in */) {
   double* vec = NULL;
   int i,local_ok = 1;
```

```
   if (my_rank == 0) {
      vec = malloc(n*sizeof(double));
      if (vec == NULL) local_ok = 0;
      Check_for_error(local_ok,"Print_vector",
            "Can't allocate temporary vector",comm);
      MPI_Gather(local_vec,local_n,MPI_DOUBLE,
            vec,local_n,MPI_DOUBLE,0,comm);
      printf("\nThe vector %s\n",title);
      for (i = 0; i < n; i++)
         printf("%f ",vec[i]);
      printf("\n");
      free(vec);
   }  else {
      Check_for_error(local_ok,"Print_vector",
            "Can't allocate temporary vector",comm);
      MPI_Gather(local_vec,local_n,MPI_DOUBLE,
            vec,local_n,MPI_DOUBLE,0,comm);
   }
}  /* Print_vector */
```

Örn. MPI ile Matris-Vektör Çarpımı

```
/*-----------------------------------------------------
 * Function:  Mat_vect_mult
 * Purpose:   Multiply a matrix A by a vector x.  The matrix
 *                 is distributed by block rows and the
 *                vectors are  distributed by blocks
 * In args:   local_A:  calling process' rows of matrix A
 *     local_x:  calling process' components of vector x
 *     local_m:  calling process' number of rows
 *     n:        global (and local) number of columns
 *     local_n:  calling process' number of components of x
 *     comm:     communicator containing all
 *                           calling processes
 * Errors:    if malloc of local storage on any process fails,
 *            all processes quit.
 * Notes:
 * 1.  comm should be MPI_COMM_WORLD because of call
 *         to Check_for_errors
 * 2.  local_m and local_n should be the same on all
 *         the processes
 */
```

```
void Mat_vect_mult(
      double    local_A[]  /* in  */,
      double    local_x[]  /* in  */,
      double    local_y[]  /* out */,
      int       local_m    /* in  */,
      int       n          /* in  */,
      int       local_n    /* in  */,
      MPI_Comm  comm       /* in  */) {
   double* x;
   int local_i,j;
   int local_ok = 1;

   x = malloc(n*sizeof(double));
   if (x == NULL) local_ok = 0;
   Check_for_error(local_ok,"Mat_vect_mult",
         "Can't allocate temporary vector",comm);
   MPI_Allgather(local_x,local_n,MPI_DOUBLE,
         x,local_n,MPI_DOUBLE,comm);

   for (local_i = 0; local_i < local_m; local_i++) {
      local_y[local_i] = 0.0;
      for (j = 0; j < n; j++)
         local_y[local_i] += local_A[local_i*n+j]*x[j];
   }
   free(x);
}  /* Mat_vect_mult */
```

Teşekkürler.

Sorular...